

Hardware Interrupts

1. Introduction

Hardware interrupts are signals sent to the microprocessor from external devices. There are two primary hardware interrupt inputs: **NMI** and **INTR**.

1.1 NMI (Non-Maskable Interrupt)

Concept

The **NMI** is a non-maskable hardware interrupt that always triggers a **Type 2** interrupt when activated. It is often used for critical system events like power failure.

Operation/Rule

Triggering and Timing Requirements:

- **Edge-Triggered:** NMI is positive edge-triggered (0-to-1 transition).
- **Pre-trigger State:** The NMI pin must be at logic 0 for at least **2 clock periods** before the positive edge occurs.
- **Hold Requirement:** Once the positive edge occurs, the NMI pin must remain at logic 1 until it is recognized by the microprocessor.

1.2 INTR (Interrupt Request)

Concept

The **INTR** input is a maskable interrupt. Unlike NMI, it is **level-sensitive** and its behavior is controlled by the Interrupt Flag (IF).

Technical Detail

Vector Assignments:

- **Flexible Range:** Any interrupt vector type number (typically in the range **20H to FFH**) can be assigned to an INTR input.
- **Reserved Vectors:** Vector numbers **00H through 1FH** are reserved for internal and software interrupts.

Operation/Rule

Operational Behavior:

- **Level-Sensitive:** Triggered by a logic high level from an external event.
- **Auto-Clear:** The INTR input is automatically disabled or cleared once the microprocessor accepts the interrupt.

2. Interrupt Acknowledgement Cycle ($\overline{\text{INTA}}$)

When the microprocessor accepts an INTR, it enters the $\overline{\text{INTA}}$ (**Interrupt Acknowledge**) cycle to fetch the interrupt vector number from the external hardware.

Concept

$\overline{\text{INTA}}$ Signal: An active-low signal used to notify external hardware to apply the interrupt vector type number to the data bus connections (D_7 – D_0).

2.1 Hardware Implementation (80H Example)

Technical Detail

Circuit Logic: A common implementation uses a **74ALS244** three-state octal buffer to place the vector number on the bus during the $\overline{\text{INTA}}$ cycle.

- **Buffer Control:** The $\overline{\text{INTA}}$ signal is connected to the buffer's enable pins (1G and 2G). A low $\overline{\text{INTA}}$ enables the buffer.
- **Vector Setting:** Switches and pull-up resistors (10K) define the bits.
- **Logic:** Closing a switch grounds the input (Logic 0); opening it pulls it high (Logic 1).

Operation/Rule

Example: Generating Vector 80H (Binary 1000 0000) To provide vector 80H to the data bus:

1. The switch for bit D_7 is kept **open** (Logic 1).
2. The switches for bits D_6 through D_0 are **closed** (Logic 0).

During the $\overline{\text{INTA}}$ cycle, the buffer drives '1000 0000' onto the low data bus.